ments even under harsh environmental conditions. TSN technology provides accurate synchronisation over long distances and minimises cabling costs. Altitude, creap, hass halt, salt fog, acid, drop, bump and ingress protection are some test features required for rugged applications.

There must be no temperature drift over time and stability. Devices should be able to withstand a temperature range of -40°C to 85°C, 100 grams of shock and 10 grams of vibration. Other integrated features include robust packaging, software for integration into test platforms, robust data logging and analytics, customisable according to customer-specific applications and suitability for OEM applications.

There are many design techniques used for achieving ruggedisation, such as use of conformal coatings and epoxies, rugged connectors, matching of thermal coefficients and so on. For testing, bake-and-shake tables must have high gravitational force.

Parameters to be measured

The parameters that need to be tested in rugged applications are as follows:

Temperature. Rugged applications often include testing in extreme temperature ranges, which can be a constraint on hardware. For example, cold start engine testing uses a test cell that can drop to -30°C. Placing hardware that is not built to withstand this range can cause components within the hardware to work incorrectly and result in incorrect data.

Shock and vibration. Monitoring the main gearbox of a bucket-wheel excavator power transmission system inside a Formula SAE race car requires consideration of vibration and shock values. If the hardware is placed where high vibration or shock values occur, it will not be able to handle these values. This can damage hardware components.

Environmental certifications. It is important to consider the environment where the tests are being conducted, especially in places like chemical factories and refineries.

Form factor. The test system's footprint is significant while deciding where the hardware will be used, such as on deck an offshore oil rig or in the middle of a desert. If the hardware has a large footprint, it can limit where tests can be conducted.

Fig. 2: FieldDAQ (Credit: www.ni.com)

In addition, designers should consider how the hardware cools itself. A hardware with no moving parts in it cools passively. In an actively-cooled device, there are energy-consuming mechanical components for other rugged considerations, which can limit where tests can be conducted.

Certifications required

Certifications for operating in hazardous locations depend on the location of the test. One should run the entire test setup to ensure it is compliant to be used in harsh locations. UL certification is required for products to operate in harsh conditions. It comes in a degree of classes and divisions. Division indicates condition and classes indicate type of hazardous location.

Another certification for rugged applications is Lloyd's Register Type Approval. It applies to products used in marine and offshore applications, and industrial plants and processes. This ensures that product performance is maintained in marine environmental conditions. Hardware must go through an entire process laid out by a third party to obtain this certification.

Ingress Protection Marking (IP rating) classifies the degree of protection provided against intrusion of dust and water. There are two numbers in IP rating of hardware. The first number indicates the level of protection against solid objects such as dust or moving parts, and ranges from zero to six. The second number indicates the level of protection against ingress of water, and ranges from zero to nine. Each number signifies a different level of protection.

Instruments for rugged environments

Ajay Singha, principal engineer, Lambda Consulting, opines that agriculture, space, industrial and defence are the key areas for the test and measurement of rugged applications. He believes that rugged test parameters go beyond commercial temperature limits. These also test products for high vibration and high humidity—conditions that may exist in rugged environments. Most of these tests usually involve highly accelerated stress testing methodologies.

Supporting Singha's view, Marc Dore, vice president - sales, Rugged Monitoring Inc., says, "Food and beverage processes, industrial process control and monitoring in chemical and process industries, medical equipment testing (MRI, pet scan, NMR, etc), electric vehicles and battery testing, and wood drying industry are other rugged applications that require accurate testing solutions. All rugged monitors must have NIST calibration certification, and be battery-operated for electrical isolation."

FieldDAQ provided by National Instrument simplifies distributed measurements with LabVIEW, to visualise and analyse real-world signals to make data-driven decisions. LabVIEW simplifies automation and customisation with seamless hardware integration, approachable programming and built-in analysis algorithms. Some coaxial connectors and cable assemblies used for this application are SMA, DIN, NEX10, BNC, SMB, UFL, IPEX and MMCX.

With many latest offerings flooding the market, one might find it difficult to decide which instrument to use. As a precautionary measure, engineers can check the type of environment for the application and how additional external factors can affect the setup. This will help them determine whether to develop an enclosure for the system or build select hardware for ruggedness. EFY